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SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0506- COMPUTER VISION

MODEL LAB PRACTICALS

SET 1

1. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to Blur using Gaussian Blur
2. Perform basic video processing operations on the captured video. Read captured video python and display the video, in slow motion and in fast motion.
3. Design and implement a water marking technique to insert the watermark into the original effectively image using Open CV.
4. Implement the Top hat technique as a Morphological operation to dilate the foreground regions based on Open CV.

Viva attached

12/100

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ITA0506

Computer
vision

MODEL LAB

- 1) Aim : To perform basic image handling and processing operations on the image and to convert Blue using Gaussian Blur.

Procedure :

- * open the new file in the idle.
- * Now import the required libraries.
- * Now import the cv
- * Now take one image on the input.
- * Now implement the code to convert blue using Gaussian Blur.
- * Now run the program.

Program :

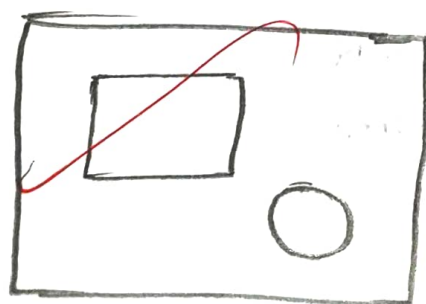
```
import cv2
img = cv2.imread('input.jpg')
if img is None:
    print ("Image Not found! check path")
    exit()
print ("Shape : ", img.shape, "size : ", img.size)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```

resized = W2.resize (image, (500,500))
blurred = W2.GaussianBlur (img, (15,15), 0)
W2.imshow ('original', img)
W2.imshow ('Gray', gray)
W2.imshow ('Gaussian Blur', blurred)
W2.imwrite ('blur - Output.jpg', blurred)
W2.waitKey(0)
W2.destroyAllWindows()

```

Input :



Output

Original
image

Output:



Blur image

Result: Program executed successfully

2) Aim: To perform basic video processing operations on the captured video and to display it on slow and fast motion.

Procedure:

* Open the new file in the IDE.

* Now import the required libraries.

* Now import the cv.

* Now take one ~~image~~^{video} on the input.

* Now implement the code to ~~can~~ display the video in slow motion and fast motion.

* Now run the program.

Program:

```
import cv2
```

```
cap = cv2.VideoCapture(0)
```

```
while True:
```

```
    ret, frame = cap.read()
```

```
    if not ret: break
```

```
    cv2.imshow('video', frame)
```

```
    k = cv2.waitKey(5)
```

```
    if k == 27: break
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```


Output:

* Normal: cv2.waitKey(30) - 30 FPS

* Slow Motion: cv2.waitKey(100) +

Save at 10 FPS $\rightarrow$ video plays 2x slower

* Fast motion: cv2.waitKey(5) + Save at

60 FPS $\rightarrow$ video ~~plays~~ 2x faster

Result: Code has been executed successfully.

3) Aim: To Design and implement a water marking technique to insert the watermark into the original effectively image using open cv.

Procedure:

* open the new file in the idle.

* Now import the required libraries.

* Now import the cv.

* Now take one image on the input.

* Now implement the code to design and implement a water marking technique to insert water mark in the original image.

Code:

```
import cv2
```

```
img = cv2.imread('input.jpg')
```

```
wm = cv2.putText(img.copy(), "SIMATS" (400, 450),
```

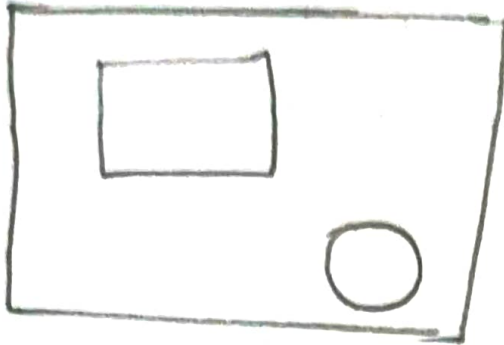
```
cv2.FONT_HERSHEY_SIMPLEX,
```

```
blend = cv2.addWeighted(img, 0.6, wm, 0.4, 0)
```

```
cv2.imshow('watermarked', blend)
```

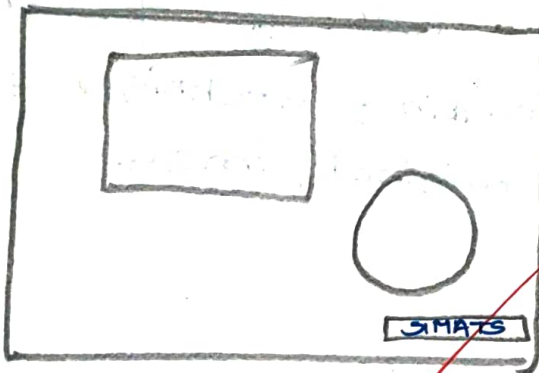
```
cv2.waitKey(0)
```

Input:



original
image.

Output:



Watermarked
image

Result: Program has been executed
Successfully.

4. Aim: To implement the Top hat technique as a Morphological operation to dilate the foreground regions based on open cv.
Procedure:

- * Open the new file in the idle
- * Now import the required libraries.
- * Now import the cv.
- * Now take one image on the input
- * Now implement the code to implement the Top hat technique as a Morphological operation.

Code:

```
import cv2
import numpy as np
img = cv2.imread('input.jpg', 0)
if img is None:
    print("Image not found")
    exit()
```

```
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (15, 15))
tophat = cv2.morphologyEx(img, cv2.MORPH_TOPHAT, kernel)
dilated_tophat = cv2.dilate(tophat, kernel, iterations=1)
cv2.imshow('original gray', img)
```



```
w2.imshow('Top Hat', tophat)
```

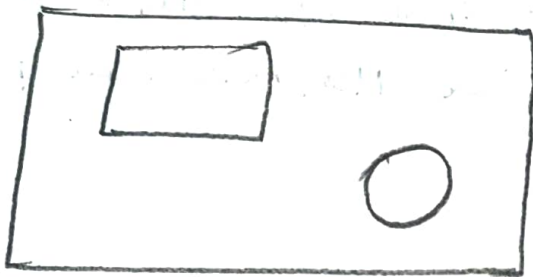
```
w2.imshow('Dilated Foreground (Top Hat)',  
          dilated - tophat)
```

```
w2.imshow(dilated tophat - output.jpg', tophat)
```

```
w2.waitKey(0)
```

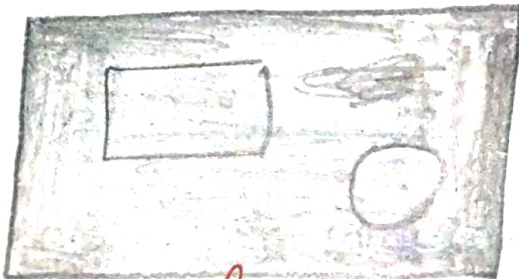
```
w2.destroyAllWindows()
```

Input:



Original
image

Output:



Gray - Scale
image

Result: The ~~program~~ program has been executed
Successfully.